Question	Correct Answer	Justification of the Correct Answer
1	C	The Physical Layer (Layer 1) deals with the physical medium, connectors, voltage levels, and the actual transmission of raw bits, without any regard for structure or meaning.
2	D	The Application Layer in the TCP/IP model is responsible for user-facing services and protocols, which encompasses the functionality of the OSI Application, Presentation, and Session layers.
3	C	CSMA/CD is a contention method where a device checks if the channel is busy (Carrier Sense) and stops transmission if a simultaneous transmission (Collision Detection) is noticed, minimizing data corruption on shared media.
4	B	CAT5e (Category 5 enhanced) was developed to improve upon CAT5 by introducing stricter specifications for reducing crosstalk (signal interference between wire pairs), which allows it to support 1000Mbps speeds reliably.
5	D	UDP (User Datagram Protocol) is a simple, connectionless transport protocol. It sends packets (datagrams) without first establishing a session, making it fast but unreliable as it lacks error recovery and sequencing mechanisms.
6	C	A Class C IP address has a default subnet mask of 255.255.255.0. In binary, this is 24 consecutive 1s, meaning the first 24 bits define the network portion.
7	C	The IOS (Internetworking Operating System) CLI is the command-line interface software that users directly interact with to configure, manage, and troubleshoot Cisco routers and switches.
8	C	A VLAN works by segmenting a network at Layer 2 into smaller, isolated broadcast domains. Traffic from one VLAN cannot be received by devices on another VLAN without the intervention of a Layer 3 device (router/L3 switch).
9	D	DAS (Direct Attached Storage) refers to storage devices (like internal or external HDDs/SSDs) that are connected directly to the server host via interfaces like SATA, SAS, or USB, and are not shared across a network.
10	B	Ceph is a prominent example of a software-defined storage platform that supports object, block, and file storage, and is often deployed as a high-performance distributed/parallel file system in large clusters.
11	B	The Network Layer (Layer 3) is where logical addressing (IP addresses) and routing decisions are made, enabling packets to traverse multiple interconnected networks.
12	C	STP (Shielded Twisted Pair) includes a metallic shield (foil or braid) layer surrounding the twisted wire pairs, which provides greater immunity to Electromagnetic Interference (EMI) and Radio Frequency Interference (RFI).
13	A	For a /24 subnet mask (255.255.255.0), the first three octets

		(192.168.5) are fixed as the Network portion. The last octet is the host portion. The Network Address is always the first address in the range, which is 192.168.5.0.
14	B	A subnet mask of 255.255.255.224 corresponds to a /27 prefix length, which uses 5 host bits. This allows for 25−2=30 usable hosts. A /26 (255.255.255.192) uses 6 host bits, supporting 26−2=62 usable hosts, which is the smallest block size that meets the 52 host requirement.
15	C	Distance Vector protocols determine the best path based on a metric (distance) and a vector (the next-hop router) provided by their direct neighbors, not by having a full topology map.
16	C	In OSPF, Area ID 0 is designated as the Backbone Area or main connection hub for all other non-backbone areas, forming the hierarchical structure of the protocol.
17	B	The default priority number for a switch running STP is 32768. This value is combined with the switch's MAC address to form the Bridge ID, used in the Root Bridge election.
18	A	The Root Port is the switch port that provides the lowest cumulative cost (path cost) from a non-root bridge to the Root Bridge. Every non-root bridge must have exactly one Root Port.
19	B	IEEE 802.1Q is the international standard for tagging Ethernet frames with VLAN identification on trunk links, ensuring frames are delivered to the correct VLAN across multiple switches.
20	C	A VTP Server switch is the only mode that can create, modify, and delete VLANs, and it sends out VTP advertisements to notify other switches of these changes, updating the entire VTP domain.
21	C	Since VLANs segment broadcast domains (Layer 2), communication between different VLANs requires a Layer 3 function: routing, which is performed by a Router or a Layer 3 Switch.
22	B	NAT Overloading (also known as Port Address Translation or PAT) allows multiple private IP addresses to be mapped to a single public IP address using different port numbers to keep traffic streams separate.
23	C	An IPv6 address is 128 bits long. It is typically split into a 64-bit network prefix and a 64-bit Interface ID.
24	B	The Configuration Register (stored in NVRAM) is a 16-bit software register whose value determines how the router boots (e.g., from ROMMON, Flash, or what configuration file to load).
25	B	EIGRP's metric calculation uniquely incorporates Bandwidth and Delay (by default) to create its composite metric (Feasible Distance), giving it more intelligence about link quality than simple hop-count protocols like RIP.
26	D	In InfiniBand, a Queue Pair (QP) represents the end-to-end communication channel. It consists of a Send Queue and a Receive Queue, defining the logical connection between two endpoints.
27	C	RAID 5 uses striping across disks and distributes parity information across all drives. This provides high read/write performance and fault tolerance against a single disk failure.

28	C	NFS (Network File System) is a distributed file system protocol that allows a user on a client computer to access files over a computer network much like local storage is accessed.
29	B	NAS is file-level storage, presenting storage resources to clients as a network share path (e.g., via NFS or CIFS). SAN is block-level storage, presenting storage to servers as raw disk volumes (blocks) over protocols like Fibre Channel or iSCSI.
30	B	The Object Storage Target (OST) is the component in the Lustre architecture that stores the actual file data. The Metadata Target (MDT) stores the directory structure and file attributes.
31	B	The required mask is /26 (block size 64). Subnet 1 is 192.168.1.0. Subnet 2 is 192.168.1.64. The first usable host in Subnet 2 is the network address +1, which is 192.168.1.65.
32	B	Stuck-in-Active (SIA) is a state where an EIGRP router has sent a query for a route (it is "Active") but has not received a reply from all neighbors in the required time, indicating a potential neighbor or network failure.
33	C	The Bridge ID is the tuple (Bridge Priority, MAC Address). If two switches share the same default or configured priority, the switch with the lowest MAC address is elected as the Root Bridge.
34	C	A VTP Transparent switch ignores all VTP advertisements for its own database, but it will forward VTP frames received on trunk ports to other trunk ports, effectively relaying advertisements through its domain.
35	C	InfiniBand and OmniPath use a credit-based flow control mechanism at the link layer. Before a sender transmits a packet, it must ensure the receiver has enough available buffer space (credits) to store the packet, preventing buffer overflow and packet loss.
36	D	RAID 0 uses disk striping without any redundancy, resulting in the highest possible read/write speed (especially for large, sequential data) but if any single disk fails, all data is lost.
37	B	GPFS (now IBM Spectrum Scale) is specifically designed to provide high-performance, concurrent access to the same file data from multiple nodes in a clustered environment, essential for large-scale data sharing.
38	A	In IPv6, a /48 prefix (first 48 bits) is the Global Routing Prefix. The next 16 bits (49 to 64) are the Subnet ID. Inserting the subnet ID 0001 into the designated 16 bits gives 2001:0db8:1234:0001::/64.
39	B	SSDs use flash memory and electronics with no mechanical parts. HDDs must wait for platters to spin and the head to physically move (seek time), which is the primary source of their higher random access latency.
40	B	The Presentation Layer (Layer 6) is responsible for data translation, formatting, compression, and encryption/decryption, ensuring that the information received from the Application layer is readable by the receiving system.

>>1:

• Internetworking

• OSI model
• Ethernet
• Wireless Networking
Assignment:
• Difference between UTP & STP
• Write categories of cables
• e in CAT5e
• OSI Model
>>2:
• Internet Protocol
• TCP/IP model
Assignment:
• IP
• TCP/IP Model
• Write a difference between TCP & UDP.
>>3:
• IP Subnetting & Variable Length Subnet Masking
Assignment:
• Subnetting
• Determine the network and host part of 192.168.5.85 /24, 10.128.240.50/30 address.
• Divide the network as per the following requirement: 192.168.1.0/24
o Subnet 1 =28 hosts
o Subnet 2 =52 hosts
o Subnet 3 =15 hosts
o Subnet4 = 5 hosts

>>4:

• Router IOS & Security Device Manager
Assignment:
• Subnet the Class C IP Address 195.1.1.0 So that you have 10 subnets each with a maximum
12 hosts on each subnet. List the Address on host 1 on subnet 0,1,2,3,10.
• Divide the network to find 500 hosts in each subnet. 152.152.0.0 /16
>>5:
• Managing an Internetworking Router
Lab Assignments:
• Working with Router Booting, configuration registers, Router IOS, Telnet, Resolving Hostname, and Debugging.
>>6:
• Implementing Routing Protocols
Lab Assignments:
• Implementation of Static Routing, RIP, IGRP, EIGRP, OSPF
>>7:
• Layer 2 switching
Theory Assignment:
• Spanning Tree Protocol (STP)?
• Write types of Spanning Tree Protocol (STP).
• priority number of Spanning Tree Protocol (STP)
• disable Spanning Tree Protocol (STP)

Lab Assignments:
• Configuration of switch, STP
• Create the following topology for STP and configure a specific switch as root bridge by changing priority number.
>>8:
• Virtual LANs
Lab Assignments:
• Configure, verify, and troubleshoot VLANs (normal/extended range) spanning multiple switches, inter-VLAN routing
• Create the following topology where Switch 1 is Server mode and switch 2,3,4,5 are client
mode. Use network 192.168.10.0 /24
o Transfer the VLAN info from the server to all clients.
o Assign a VLAN as mention in the topology.
o PC's in having same VLAN should ping each other.
• Create the following topology where Switch 1 is Server mode and Switches 2,4, are client mode. Use network 192.168.1.0 /24
o Transfer the VLAN info from the server to all clients.
o Assign a VLAN as mention in the topology.
o PC's in having same VLAN should ping each other.
o Configured switch 3 in transfer mode and transferred vlan information to switch 4

>>9:
• NAT
• IPV6
Lab Assignments:
• NAT configuration, Configuring Routers with IPv6.
>>10:
• OmniPath Architecture
• Omnipath Packet Structure
• Infiniband Architecture
• Infiniband Packet Structure
>>11:
• Storage Devices
• Types of HDD
• RAID and RAID Levels
• File Systems(Windows and Linux)
• Network File System.
• DAS/NAS/SAN
>>12:
Parallel File System
• Introduction
• Classification
• Ceph File System Architecture and Components
• GPFS features and architecture.
• Lustre File System Architecture and Components